Problem Statement - RidePulse

Modern cities are facing growing transportation problems such as heavy traffic, unsafe driving, frequent accidents, and slow response to road incidents. Most existing systems rely on manual processes and don't use real time data effectively, making them reactive rather than proactive.

Drivers usually do not receive feedback on their driving behavior, while passengers often have limited and inefficient ways to report unsafe situations or road issues. At the same time, authorities struggle to monitor traffic conditions accurately and respond quickly due to a lack of centralized and reliable data.

In the public transport sector, bus owners face significant challenges due to the lack of proper monitoring and control over their vehicles and staff, which can lead to misuse, fraud, or irresponsible behavior that affects both service quality and revenue. Additionally, there is no structured welfare or support system for drivers and conductors, when they face accidents or unexpected issues, they are often left without assistance or protection, making them vulnerable and financially unstable. Another major limitation is the absence of predictive systems that can identify potential risks before they occur.

As a result, road safety decreases, accountability is weakened, and overall transportation management becomes inefficient. To overcome these challenges, there is a clear need for a smart and integrated eco system that connects all stakeholders, collects real-time data, and provides meaningful insights. The RidePulse application is proposed to address this problem by offering a centralized, data driven platform that monitors driving behavior, enables easy complaint reporting, improves transparency for bus owners, and uses predictive analytics to enhance road safety, accountability, and support better decision making.